

Exercise 6

For $z \in \mathbb{R}^n$, write $\hat{F}_z(v) = \frac{1}{n} \sum_{i=1}^n \chi_{\{z_i \leq v\}}$ for the empirical cumulative distribution function of z and $\text{Quantile}(z; \tau) = z_{(\lceil \tau n \rceil)}$ for any $\tau \in (0, 1]$. $\text{Quantile}(z; 0)$ is defined as $-\infty$.

Prove that, for any $z \in \mathbb{R}^n$, the following statements are equivalent:

1. $\hat{F}_z(v) = \sup \{ \tau \in (0, 1) \mid \text{Quantile}(z; \tau) \leq v \}$ for all $v \in \mathbb{R}$.
2. $\text{Quantile}(z; \hat{F}_z(v)) \leq v$ for all $v \in \mathbb{R}$.
3. $\hat{F}_z(\text{Quantile}(z; \tau)) \geq \tau$ for all $\tau \in [0, 1]$.

Prove also that, in case z has no repeated values, all above statements are equivalent to the statement that, for all $\tau \in [0, 1]$,

$$\hat{F}_z(\text{Quantile}(z; \tau)) = \frac{\lceil \tau n \rceil}{n}.$$

Solution NEED WORK ///